VÃYU

COMMITTED TO LIFE

Vayu Products Private Limited

www.vayuproducts.com

About VAYU

Vayu Products Private Limited

As North India's leading manufacturer of liquid gases, Vayu Products Private Limited delivers over 400 tonnes per day, making us one of the largest producers of Liquid Oxygen, Liquid Nitrogen, Liquid Argon, and Liquid Medical Oxygen. Our cutting-edge facilities in Haryana and Punjab ensure consistent, high-quality supply to industries and hospitals, meeting the region's growing demand for these essential gases.

Vayu Speciality Gases Private Limited

A sister concern of Vayu Products, Vayu Speciality Gases is dedicated to supplying speciality gases tailored to India's booming semiconductor and photovoltaic industries. With a state-of-the-art storage facility and a fully integrated supply chain, we offer a complete solution for all speciality gas needs, positioning VAYU as your trusted partner in this high-growth sector.

VAYU
COMMITTED TO LIFE

Our Services

Liquid Nitrogen (LN2)

Essential for cryogenic freezing, industrial cooling, and used in semiconductor and photovoltaic manufacturing for its ultra-low temperatures and inert properties.

Liquid Oxygen (LOX)

Widely used in metal cutting, welding, industrial oxidation, and supporting high-tech manufacturing processes in the semiconductor industry.

Liquid Argon (LAR)

A key inert gas used in semiconductor production, welding, and lighting, providing a stable atmosphere for precise manufacturing processes.

Liquid Medical Oxygen (LMO)

Vital for healthcare applications, ensuring safe and life-saving medical use.

Industries We Serve

- Semiconductors & Photovoltaics
- Healthcare
- Metal Fabrication
- Welding & Cutting
- Research & Development

VĀYU
COMMITTED TO LIFE

Why Choose
Vayu Products

01

Regional Leader

Among the top air gas manufacturers in North India.

02

Quality & Purity

All products meet the highest industry and safety standards, ensuring reliable performance.

03

Customer Focused

Tailored solutions and exceptional service for all our clients

04

Consistent Supply

A fully integrated supply chain guarantees timely and efficient delivery

05

Lifeblood of Industry

Our gases drive critical sectors like semiconductors and photovoltaics, along with healthcare and industrial manufacturing

VÃYU

COMMITTED TO LIFE

Specialty Gases for Semiconductor and Photovoltaic Applications

From Vayu Speciality Gases Private Limited

www.vayuproducts.com

VÃYU
COMMITTED TO LIFE

Key Specialty Gases We Offer

Silane (SiH_4)

Role: Key in thin-film silicon deposition, used in solar cells and semiconductors.

Applications: Thin-film solar cells, semiconductor devices, silicon-based coatings.

Features: Pyrophoric and requires inert handling.

Ammonia (NH_3)

Role: Essential for creating silicon nitride layers in anti-reflective coatings and dielectric layers.

Applications: Nitration, silicon nitride deposition.

Features: Toxic and corrosive, requiring specialised safety measures.

Nitrous Oxide (N_2O)

Role: Vital for oxide growth and silicon dioxide layer formation.

Applications: Oxide growth, PECVD, silicon oxide deposition.

Features: Non-flammable, with anaesthetic effects upon inhalation.

Phosphine (PH_3)

Role: Used for n-type doping in semiconductors, introducing phosphorus atoms.

Applications: n-type doping, LEDs, integrated circuits.

Features: Highly toxic and flammable, requiring strict handling.

Hydrogen (H_2)

Role: Used in annealing, passivation, and reducing atmospheres in semiconductor processes.

Applications: Thin-film deposition, silicon wafer annealing.

Features: Highly flammable, handled with care in inert environments.

Trimethylamine (TMA)

Role: Precursor for Atomic Layer Deposition (ALD) of aluminium oxide.

Applications: Barrier layers, high-k dielectrics, semiconductors.

Features: Pyrophoric, requiring special storage and handling.

Boron Trichloride (BCl_3)

Role: Key in thin-film silicon deposition, used in solar cells and semiconductors.

Applications: Thin-film solar cells, semiconductor devices, silicon-based coatings.

Features: Pyrophoric and requires inert handling.

Nitrogen Trifluoride (NF_3)

Role: Key in thin-film silicon deposition, used in solar cells and semiconductors.

Applications: Thin-film solar cells, semiconductor devices, silicon-based coatings.

Features: Pyrophoric and requires inert handling.

VĀYU

COMMITTED TO LIFE

Why Choose Vayu Speciality Gases?

Our facility stores all the essential gases for semiconductor and photovoltaic applications, ensuring we meet all your production needs.

We produce liquid nitrogen, liquid oxygen, liquid medical oxygen, and liquid argon, providing a reliable domestic supply of core industrial gases.

Our advanced storage solutions ensure the safe and efficient handling of volatile and rare gases, maintaining the highest safety standards.

From production to delivery, our fully integrated supply chain ensures uninterrupted availability and fast delivery times across India

We focus on reducing environmental impact by offering greener alternatives and sustainable solutions for the industries we serve.

With our expertise, advanced technology, and nationwide supply network, Vayu Speciality Gases is your dependable partner for all your semiconductor and solar manufacturing needs.

VĀYU

COMMITTED TO LIFE

Turnkey Cryogenic
Facility Design & Installation

Vayu Products provides end-to-end solutions for designing, engineering, and executing cryogenic storage and gas delivery systems. Our turnkey service ensures that all aspects of the facility — from initial concept to regulatory approvals and installation — are expertly handled, tailored to meet the specific needs of our clients.

Our system delivers efficient and safe gas supply through the following components:

Cryogenic Storage Vessel

The storage vessel is custom-sized according to the client's operational and capacity requirements, ensuring optimal storage of cryogenic liquids.

Features Includes :

Tank Pressure Gauge: To monitor pressure for safe operations.

Pressure Relief System: Prevents excess pressure buildup.

Fill Valves: For refilling cryogenic liquid.

Contents Gauge/Tracker: Provides real-time monitoring of liquid levels.

Atmospheric Vaporizers

The vaporizers convert the stored cryogenic liquid into gas by heating it to near ambient temperature. Sized for peak demand, they ensure efficient, uninterrupted gas delivery.

Pipeline Pressure Regulation & Low-Temperature Protection

After vaporization, the gas is delivered through regulated pipelines, ensuring consistent pressure and temperature. The system is designed with safety features to avoid over-pressure or low-temperature hazards, ensuring a reliable gas supply to the customer’s pipeline.

VÃYU
COMMITTED TO LIFE

Our Turnkey Solutions Include:

End-to-End Design and Engineering:

We develop a fully customized design based on client specifications, ensuring all parameters such as capacity, size, and operational requirements are met.

Execution:

We manage the entire installation and commissioning process, ensuring the facility is built to the highest standards.

Regulatory Approvals:

We handle all regulatory compliance and approvals to ensure the facility meets national and international safety and operational standards, giving you a hassle-free experience.

How the System Works:

Storage

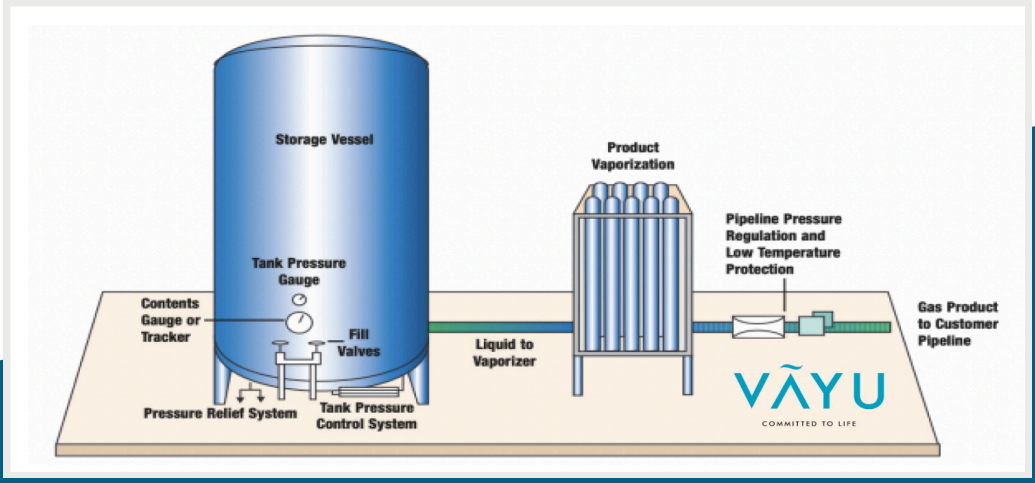
Cryogenic liquid is stored in a custom-sized vessel designed with safety features like pressure gauges, relief systems, and fill valves.

Vaporization

The liquid is directed to atmospheric vaporizers that convert it into gas for delivery.

Gas Delivery

The vaporized gas is regulated and delivered through a controlled pipeline to the customer's site, maintaining the necessary pressure and temperature.



Advanced Facilities Supporting Reliable Gas Supply

VÃYU

COMMITTED TO LIFE

Our Partners

VAYU at a Glance

- Led by promoters with over 50 years of expertise in the gases sector.
- Manufacturing capacity of 400 tonnes per day of cryogenic gases Liquid Oxygen Liquid Nitrogen, Liquid Argon and Liquid Medical Oxygen
- One of the biggest manufacturer of Liquid gases in North India

VAYU
COMMITTED TO LIFE

Contact Us

Website
www.vayuproducts.com

Email
info@vayuproducts.com

Phone No.
+91-9729265548
+91-7357270000